

Name:

Teacher:



Year 9 Stage 5.2/5.3 Mathematics Yearly Exam 2022

Instructions

- Working time: **50 minutes**
- **ALL students must attempt Section 1 and Section 2**
- Write in black or blue pen, draw diagrams, tables and graphs in pencil
- NESA Approved Calculators are allowed to be used
- Show relevant working out where necessary

Surface Area And Volume	Trigonometry	Probability	Total
/15	/15	/15	/45

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Short Answer Questions – Show working where necessary.

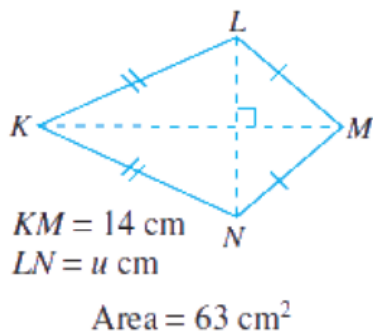
Surface Area And Volume

1. If the average file size of a photo is 2.5 MB , find the number of photos that can be stored on a 1 GB USB flash drive.

1

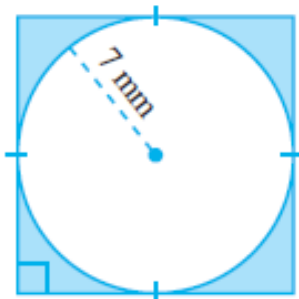
2. Find the length of the diagonal LN if the kite has an area of 63 cm^2 and the other diagonal KM has a length of 14 cm .

1



3. Find the area of the shaded part, correct to one decimal place.

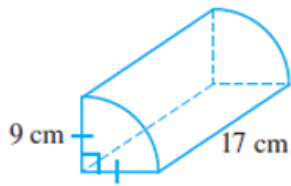
2



Section 1 continues on the following page

Short Answer Questions – Show working where necessary.

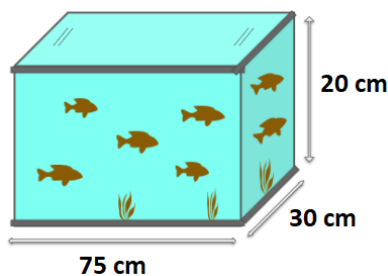
4. Find the total surface area of the solid, correct to one decimal place.

2

5. Find the side length of a cube whose surface area is 3456 cm^2 .

2

6. 28 L of water is poured into a rectangular fish tank of dimensions 75 cm by 30 cm by 20 cm . How many more litres (L) of water is required to fill the fish tank to the top?

2

Section 1 continues on the following page

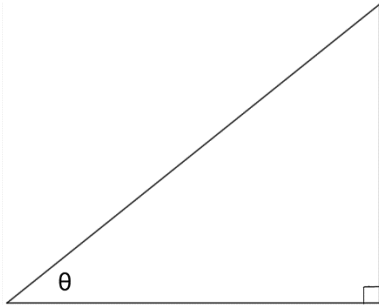
Short Answer Questions – Show working where necessary.

Trigonometry

7. A right-angled triangle has the trigonometric ratio $\tan \theta = \frac{7}{24}$.

a) Show this information on the triangle.

1



b) Find the ratio for $\sin \theta$.

2

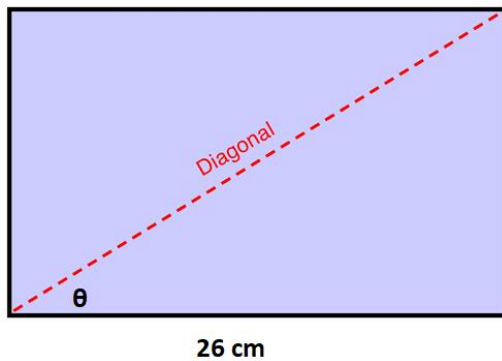
8. The entrance to a shopping centre is 60 cm above the ground level. A wheelchair ramp is built to the entrance at an angle of 12° with the ground. Calculate (in cm) the length of the ramp, correct to one decimal place.

2

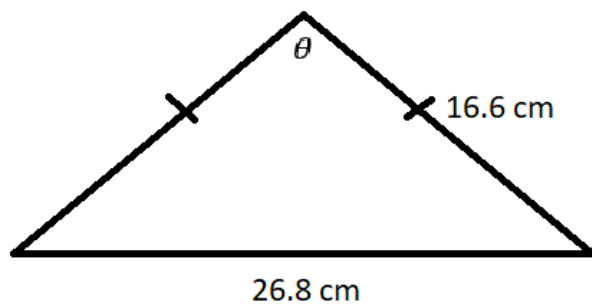
Section 1 continues on the following page

Short Answer Questions – Show working where necessary.

9. A rectangle has an area of 312 cm^2 and a length of 26 cm . Find the angle θ , formed between the length and the diagonal, correct to the nearest degree. **2**



10. Find the apex angle, θ , of an isosceles triangle if the length of the equal sides is 16.6 cm and the length of the base is 26.8 cm . Give the answer correct to the nearest minute. **2**



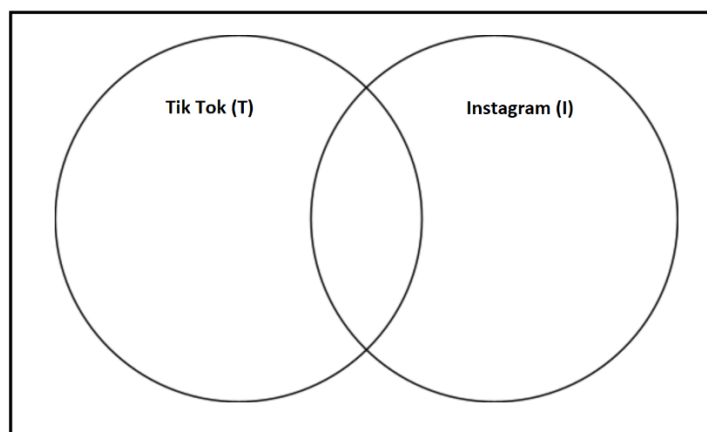
11. If $\sin \theta = 0.5432$, find $\cos \theta$, correct to two decimal places. **1**

Section 1 continues on the following page

Short Answer Questions – Show working where necessary.**Probability**

12. A survey of 55 students found that 25 students like Tik Tok (T) and 33 students like Instagram (I). 10 students like both Tik Tok and Instagram and 7 like neither Tik Tok nor Instagram.

- a) Fill in the Venn diagram to represent the survey results.

2

- b) If one of the 55 students was randomly selected, find:

i. $P(\text{likes neither Tik Tok nor Instagram})$

1

ii. $P(\text{likes Tik Tok})$

1

13. The two-way table shows data collected about Year 9 Students.

	Left-handed	Right-handed
Male	12	55
Female	10	43

- a) What is the percentage probability that a student selected at random is male and left-handed? **1**

- b) Given that a student is right-handed, what is the fraction probability that the student was also female? **1**

Short Answer Questions – Show working where necessary.

14. Chrissy is decorating the living room for a dinner party. In the box of decorations, there are 3 blue and 2 green items. Chrissy chooses between two decorations at random to put up.

- a) Draw a tree diagram showing the sample space for Chrissy selecting two decorations (without replacement) from the box.

2

- b) If the first decoration is blue, what is the fraction probability that the second decoration is green?

1

- c) What is the fraction probability of the first two decorations chosen being one of each colour (in any order)?

1**End of Section 1**

Short Answer Questions – Show working where necessary.

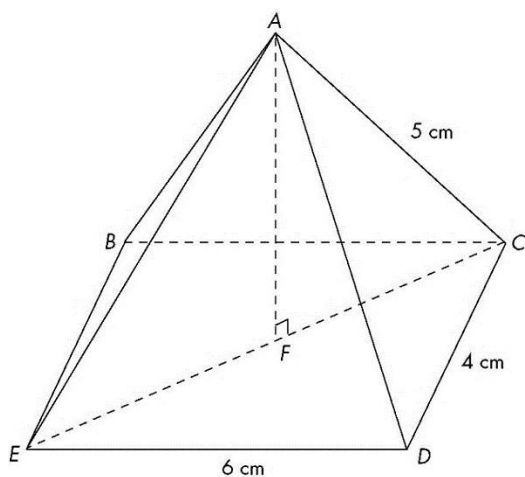
Surface Area And Volume

1. A cone has a volume of 1216.4 cm^3 . Find, correct to one decimal place, the radius of the cone if the height is 15 cm .

2

2. Calculate the volume of the rectangular pyramid, correct to two decimal places.

3

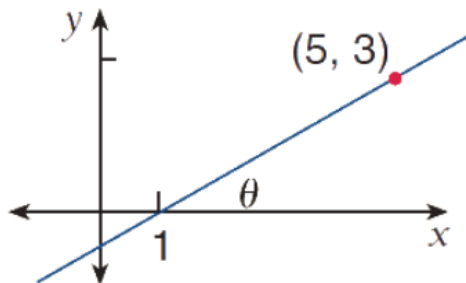


Short Answer Questions – Show working where necessary.

Trigonometry

3. Find θ , the angle made between the line and the horizontal (x) axis, correct to the nearest minute.

1



4. Two points, A and B are in line with the base of a tower 25 m high. The angle of elevation of the top of the tower is 24° from A and 38° from B . How far apart are the points A and B , correct to the nearest metre.

2

5. Gru and Kevin leave home at 7 p.m. Gru drove due north at 80 km/h while Kevin drove due east at 60 km/h . Find the bearing of Kevin from Gru at 8:15 p.m., correct to the nearest degree.

2

Section 2 continues on the following page

Short Answer Questions – Show working where necessary.**Probability**

6. One hundred citizens are surveyed regarding their use of water for their garden. 27 say that they use rain water, 42 say that they use tap water and 38 say that they do not use water at all.

a) Represent the information in the two-way table below.

2

	Rain water	No rain water	Total
Tap			
No tap water			
Total			100

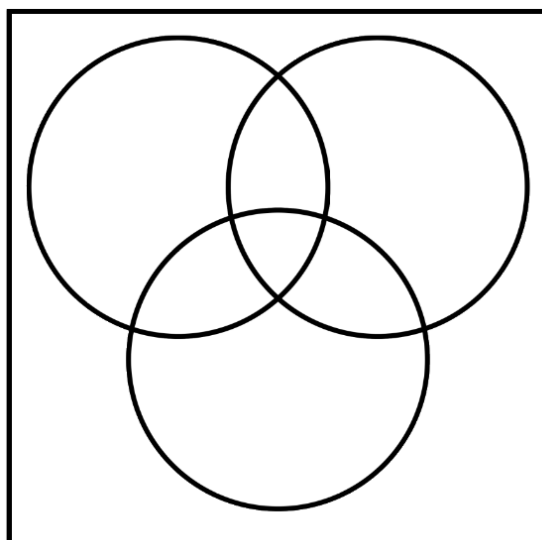
b) What is the decimal probability that one of the citizens surveyed uses tap water or rain water? **1**

7. Fill in the Venn diagram to represent the information below:

2

90 students all own a dog, a cat or a bird. They were asked which pets they own.

- 37 own a bird
- 46 own a cat
- 40 own a dog
- 16 own both a cat and a bird
- 11 own both a dog and a cat
- 12 own both a bird and a dog
- 6 own a dog, a cat and a bird



End of Exam :)